Homework 7

via Gradescope

- Failure to submit homework correctly will result in zeroes. *If you do not select the pages of your work correctly, student graders cannot see your work.*
- Handwritten homework is recommended. You do not have to type up your work.
- Problems assigned from the textbook are from the 5th edition.
- No late homework accepted. Lateness due to technical issues will not be excused.

---

1. (1 point) Use contradiction to prove the following statement

$$\forall x \in \mathbb{R},\ \text{If } |x| < \varepsilon \text{ for any } \varepsilon > 0,\ \text{then } x = 0.$$

2. (1 point) Section 4.7 #9, 18.
3. (1 point) Section 4.7 #22, 24.
4. (1 point) Section 4.7 #28. Prove this using both contradiction and the contrapositive.
5. (1 point) Section 4.7 #31.
6. (1 point) Section 4.8 #8.
7. (1 point) Prove or disprove the following conjecture.

   Conjecture. *$xy \in \mathbb{R} - \mathbb{Q}$ for any $x, y \in \mathbb{R} - \mathbb{Q}$.*

8. (1 point) Provided the Pythagorean Theorem 4.8.1, $\sqrt{2} \notin \mathbb{Q}$:
   (a) Prove that $\sqrt{6} \notin \mathbb{Q}$ using Proposition 4.7.4 and Euclid's lemma.
   (b) Prove that $\sqrt{2} + \sqrt{3} \notin \mathbb{Q}$ using the previous part.

   Proposition (4.7.4)**.** *$n^2 \in 2\mathbb{Z}$ implies $n \in 2\mathbb{Z}$ for any $n \in \mathbb{Z}$.*

   Lemma (Euclid's lemma)**.** *Let p be prime. Let $a, b \in \mathbb{Z}$ and $a > 1$ and $b > 1$. If $p \mid ab$ and $p \nmid a$, then $p \mid b$.*

9. (1 point) Section 4.10 #12, 15.
10. (1 point) Section 4.10 #22, 24.

Definition. Let $a, b \in \mathbb{Z}$, not both zero. $d \in \mathbb{Z}$ is the *greatest common divisor* of a and b, denoted $\gcd(a, b)$, if and only if

(1) $d \mid a$ and $d \mid b$

(2) For any $c \in \mathbb{Z}$, $c \mid a$ and $c \mid b$ implies $c \leq d$.

Definition. $a, b \in \mathbb{Z}$ are *relatively prime* if and only if $\gcd(a, b) = 1$.

11. (1 point) Section 4.10 #28, 29.

Definition. Let $a, b \in \mathbb{Z} - \{0\}$. $c \in \mathbb{Z}^+$ is the *least common multiple* of a and b, denoted $\text{lcm}(a, b)$, if and only if

(1) $a \mid c$ and $b \mid c$

(2) For any $m \in \mathbb{Z}^+$, $a \mid m$ and $b \mid m$ implies $c \leq m$.